MOHAMED ALAA

Senior Unity Engineer

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SUMMARY

Senior Unity Engineer with 7+ years of hands-on experience designing, building, and launching commercial 2D, 3D, Virtual Reality (VR), Augmented Reality (AR), and Multiplayer games and enterprise simulations across Android, iOS, Meta Quest, and Windows PC. Led engineering teams at Genesis Creations through July 2026, delivering cutting-edge VR training platforms and interactive 3D digital twins. Proven track record with 600,000+ combined downloads across published titles, including Experience Makkah (250,000+ downloads), Kortifo (200,000+ downloads), and Rehlatie (150,000+ downloads). Skilled in scalable C# architecture, multiplayer networking (Unity Netcode, Photon Fusion/PUN), XR mechanics (OpenXR, Meta XR SDK), performance optimization (Addressables, URP), and cloud live-ops (Firebase, Unity Gaming Services).

EXPERIENCE

Lead Unity Engineer

Genesis Creations

05/2025 - 07/2026 Cairo, Egypt

- Led and scaled an engineering team from 4 mid-level and senior Unity developers to 7+ junior, mid-level, and senior engineers across multiple concurrent VR and 3D enterprise projects.
- Architected scalable, modular C# codebases for new projects and refactored legacy systems using SOLID principles and UniTask to improve maintainability and runtime performance.
- Collaborated with Product Owners in Agile/Scrum ceremonies to define sprint goals, decompose features into actionable tasks, and achieve 100% on-time sprint delivery.
- Spearheaded company-wide development workflow upgrades by introducing Unity 6 standards, modern tooling, and strategic process optimizations.
- Mentored junior developers across multiple teams and facilitated technical bootcamp training on intermediate and advanced Unity concepts to uphold code quality standards.

Senior Game Developer

Nahdet Misr

04/2024 - 05/2025 Cairo, Egypt

- Engineered core gameplay mechanics and interactive features for Rehlatie, an educational 2D mobile game in Unity that achieved 150,000+ downloads and a 4.5+ star rating across Google Play and the App Store.
- Devised and executed a performance optimization strategy using Unity Addressables and system refactoring, reducing initial app size by 40% and improving runtime efficiency by 35%.
- Integrated Firebase backend services including Cloud Functions, Remote Config, Crashlytics, and Analytics to enable real-time remote configuration and crash monitoring.
- Managed end-to-end build and deployment pipelines for both Apple App Store and Google Play Store, ensuring full compliance with store guidelines and 99%+ crash-free release stability.
- Conducted peer code reviews and provided technical guidance to junior team members to enforce clean C# architecture and development best practices.

SKILLS

Game Engines & Rendering

Unity · Unity 6 · Unity 3D · Unity 2D · Universal Render Pipeline (URP) · Addressables System · Custom HLSL Shaders · Cinemachine · DOTween · UI Toolkit · uGUI · TextMesh Pro · SpriteShape · Physics & Animation Systems

Programming & Architecture

C# · .NET · ASP.NET · Object-Oriented Programming (OOP) · SOLID Principles · Design Patterns · State Machines · Event-Driven Architecture · Dependency Injection (Reflex) · Asynchronous Programming (UniTask)

Multiplayer & Cloud Live-Ops

Unity Netcode for GameObjects · Photon Fusion · Photon PUN · Firebase Cloud Functions · Firebase Remote Config · Firebase Crashlytics & Analytics · Unity Cloud Code & Cloud Save · In-App Purchases (Unity IAP) · RESTful APIs

AR / VR & XR Technologies

Meta Quest / Oculus SDK · Meta Interaction SDK · OpenXR · XR Interaction Toolkit · Hand Tracking · Vuforia · AR Foundation · Google Cardboard SDK · Spatial Audio

Tools, Databases & Leadership

Git & GitHub · CI/CD Pipelines · Figma · Postman · MySQL · PostgreSQL · Odin Inspector · glTFast · Jira & Trello · Agile / Scrum · Team Leadership · Code Reviews & Mentorship

Platforms

Android · iOS · Meta Quest (VR) · Windows PC

EDUCATION

Bachelor of Software Engineering

University of Ain Shams

2014 - 2018 Cairo, Egypt

- Focused on Software Engineering, Object-Oriented Design, Data Structures & Algorithms, and System Architecture.

EXPERIENCE (CONTINUED)

Senior Game Developer
UMAMI Games
10/2022 - 04/2024 Cairo, Egypt

- Developed core gameplay mechanics and backend integrations for Kortifo, a multiplayer 2D football card battle game that surpassed 200,000+ downloads and a 4.5+ star rating on App Store and Google Play.
- Implemented Unity Netcode for GameObjects to power real-time multiplayer matches, handling deterministic state synchronization and networked card actions for thousands of players.
- Designed the Daily and Weekly Objectives System using Unity Cloud Code for dynamic mission logic, alongside a deterministic Card Effect Queue System supporting 10,000+ trivia questions.
- Built a full-featured Shop System supporting virtual currency and real-money In-App Purchases (Unity IAP, Cloud Save, and Remote Config), reducing content deployment cycles by 60%.
- Translated Figma designs in collaboration with the art team into polished in-game UI/UX and supported App Store and Google Play release submissions and QA validation.

.NET Developer
EDGE PRO
03/2022 - 07/2022 Cairo, Egypt

- Constructed backend services for multiple government web platforms using ASP.NET and C#, focusing on high-availability API development and cross-system integration.
- Established secure RESTful APIs to standardize authenticated data exchange between internal services and external enterprise systems.
- Modeled relational database schemas, optimized SQL queries, and authored stored procedures in MySQL and PostgreSQL, while debugging legacy codebases to improve platform responsiveness.

Junior Game Developer
AVATARIS
01/2022 - 03/2022 Remote

- Converted Figma wireframes into responsive, interactive in-game UI systems using Unity for a cross-platform mobile game within a remote Agile team.
- Streamlined asset import workflows with graphic designers to improve mobile rendering fidelity and maintained clean, localization-ready UI architecture.

Junior Game Developer
Vhorus
10/2019 - 12/2021 Cairo, Egypt

- Directed full-lifecycle development as the sole Unity developer in a 3-person team for Experience Makkah, a cross-platform mobile VR Hajj simulation using Google Cardboard SDK that surpassed 250,000+ downloads on Android and iOS.
- Programmed scene management, interactive VR camera systems, and spatial audio while optimizing build size and rendering pipelines, reducing memory usage by 50% and sustaining 60 FPS on mid-tier mobile devices.
- Created commercial Augmented Reality (AR) brand activation apps using Vuforia and AR Foundation, and shipped Tangram of 7, a 5.0-star 2D mobile puzzle game.

Junior Game Programmer
CLOUD SOFT
02/2019 - 10/2019 Cairo, Egypt

- Delivered end-to-end 2D educational mobile games incorporating Augmented Reality (AR) features as the sole Unity developer, meeting 100% of production timelines independently.
- Coordinated asset integration, gameplay logic, UI/UX flow, and Android performance tuning to ensure stability across entry-level mobile devices.

KEY PROJECTS

VRC - Virtual Interactive Cleanroom
Team Leader & Lead Developer | Genesis Creations
2024 - 2025 Meta Quest (VR)

- Supervised 7 Unity developers building Canada's first VR sterile compounding training platform for hospital pharmacies.
- Formulated a universal dynamic surface-cleaning system, automated workflow orchestration, and real-time protocol compliance feedback.

DXB - Interactive 3D Facility Walkthrough
Lead Unity Engineer | Genesis Creations
2025 Meta Quest & PC

- Orchestrated a multi-zone 3D data center tour for GITEX demos using Meta XR SDK, OpenXR, URP, and Cinemachine.
- Connected Schneider Electric NetEco APIs via a typed UniTask layer driving live telemetry dashboards (power, cooling, alarms).

Gaming Room VR
Solo Developer | Showcase Project
2026 Meta Quest (VR)

- Simulated two-handed pool cue mechanics with cue-axis strike assist physics and angle-tested stickable darts in Unity 6.
- Calibrated custom stylized hand meshes driven directly by the Meta Interaction SDK hand skeleton across custom bone layouts.

Kortifo - Football Card Game
Senior Game Developer | UMAMI Games
2022 - 2024 Android & iOS (200K+)

- Synchronized real-time multiplayer card battles with Unity Netcode for GameObjects, Cloud Code objectives, Card Effect Queue, and IAP shop.

Experience Makkah - Virtual Hajj
Sole Unity Developer | Vhorus
2019 - 2021 Android & iOS (250K+)

- Executed full-lifecycle VR simulation with Google Cardboard SDK, spatial audio, and 60 FPS mobile rendering optimization.

Rehlatie - Educational Game
Senior Game Developer | Nahdet Misr
2024 - 2025 Android & iOS (150K+)

- Reduced initial store download size by 40% via Unity Addressables asset streaming and Firebase live-ops integration.

Darbk Khdr - Saudi Desert Adventure
Solo Developer | Cultural Heritage
2023 - 2024 Meta Quest & Mobile

- Launched complete cross-platform gameplay loop on Google Play and Meta Quest Store with desert exploration and VR locomotion.

ACTIVITIES & CERTIFICATIONS

- Instructed global learners as a Unity Developer Instructor on Coursera in intermediate/advanced Unity and C#.
- Guided aspiring developers as a Game Development Mentor at Traverse Summit 2024 (largest youth summit in MENA).
- Published reusable C# tools and game development packages as a verified Unity Asset Store Publisher.